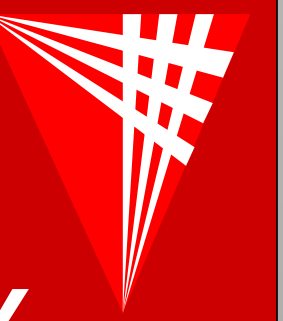


Characterizing Spectrum Usage from Long-Term Passive Measurements

Naicheng Wei, Cynthia Hood.

Department of Computer Science

ILLINOIS INSTITUTE
OF TECHNOLOGY



Spectrum Measurements

Beyond Occupancy

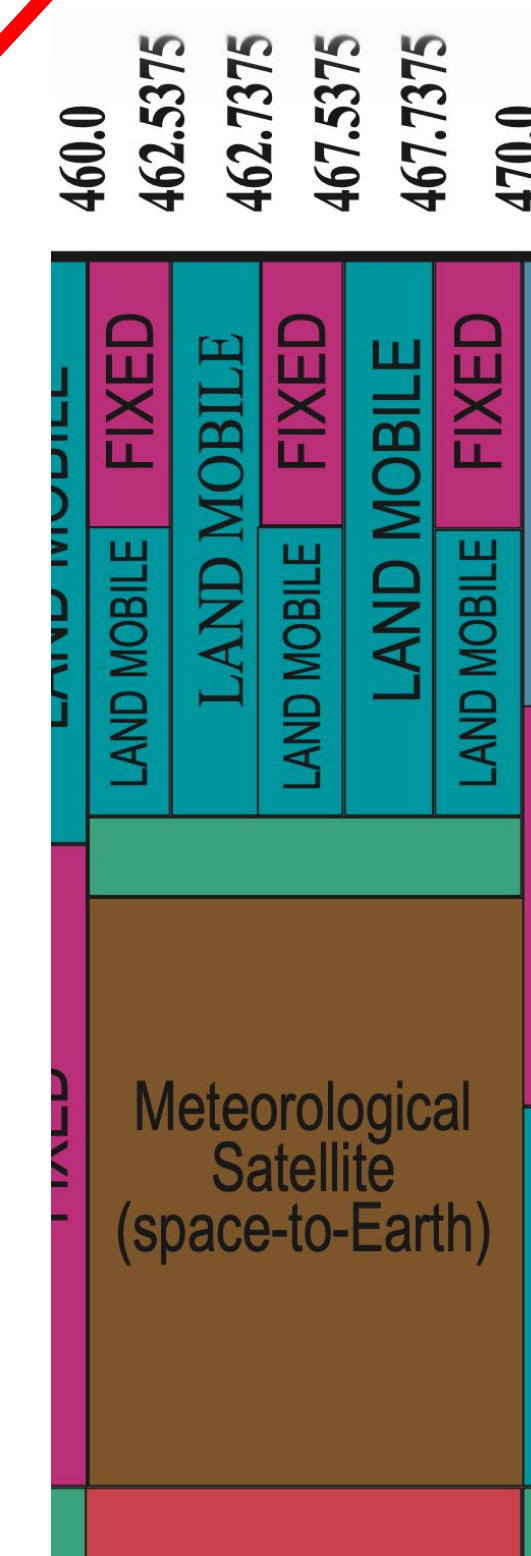
Behavioral Patterns



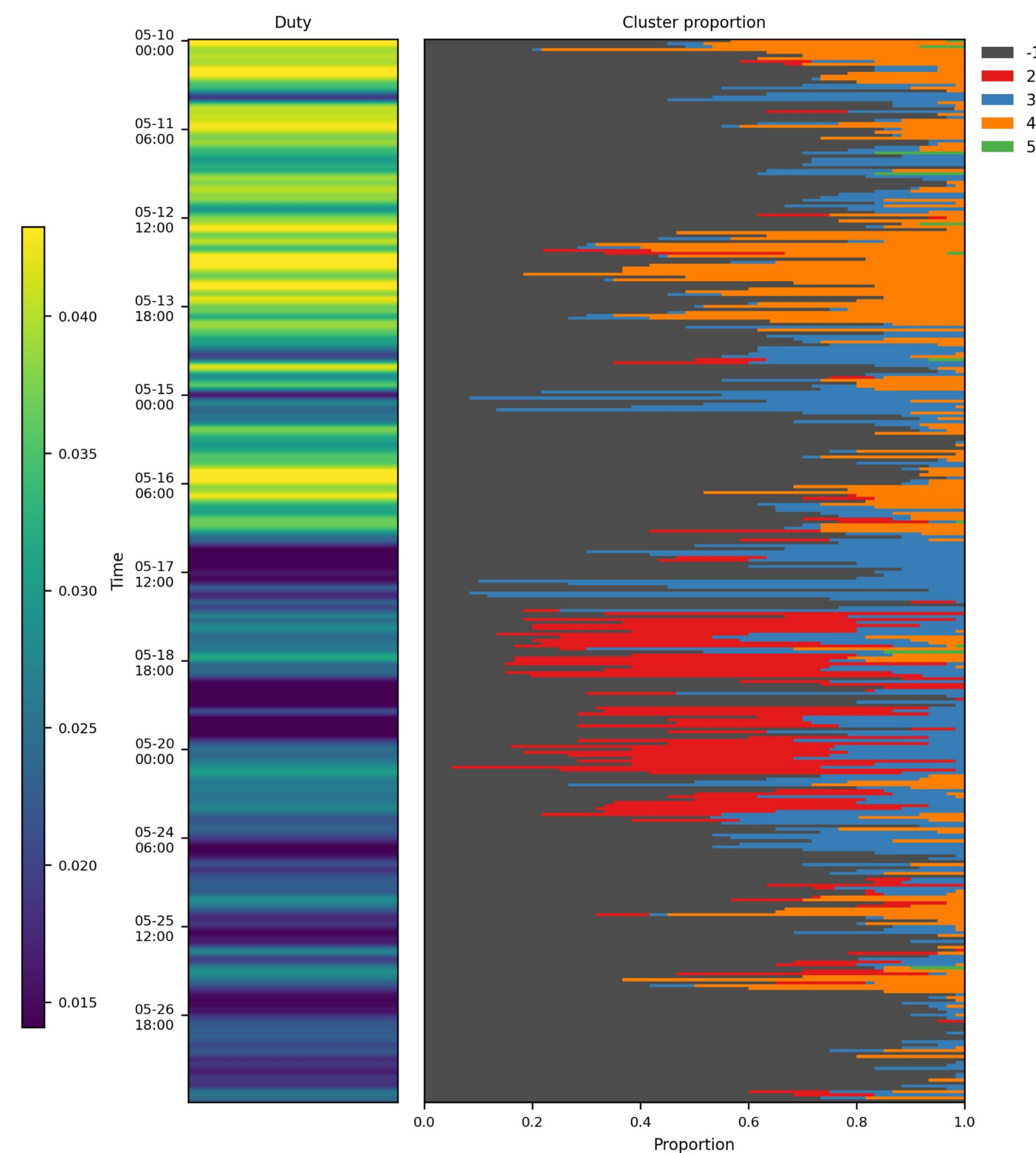
IIT Spectrum Observatory:

- Real data
- No labels
- Long-term observation
- Passive wideband power measurements (30 MHz – 6 GHz)
- Continuous monitoring since 2007

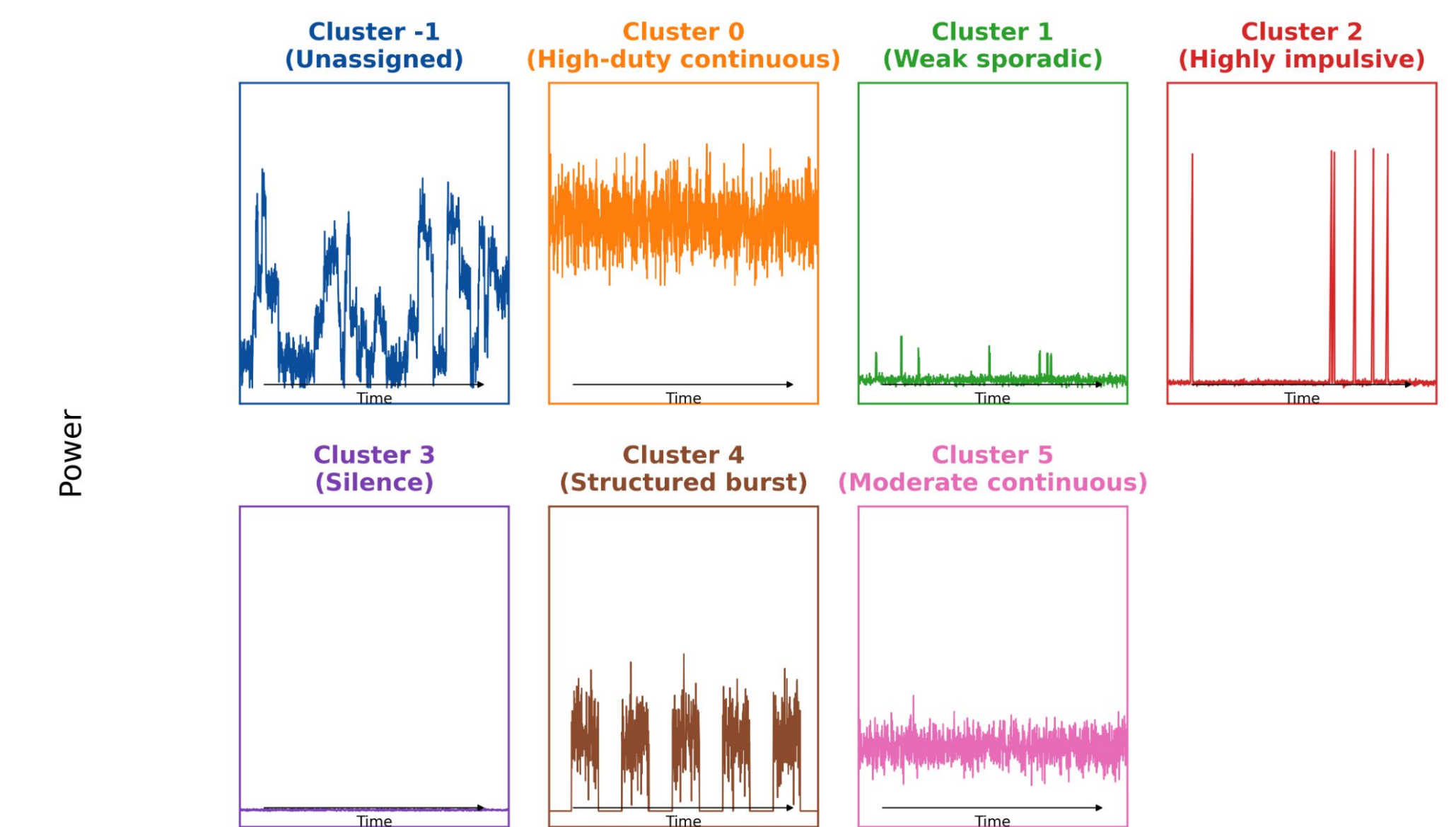
UNITED STATES
FREQUENCY
ALLOCATIONS
THE RADIO SPECTRUM



Duty vs Regime Mixture Over 16 Days

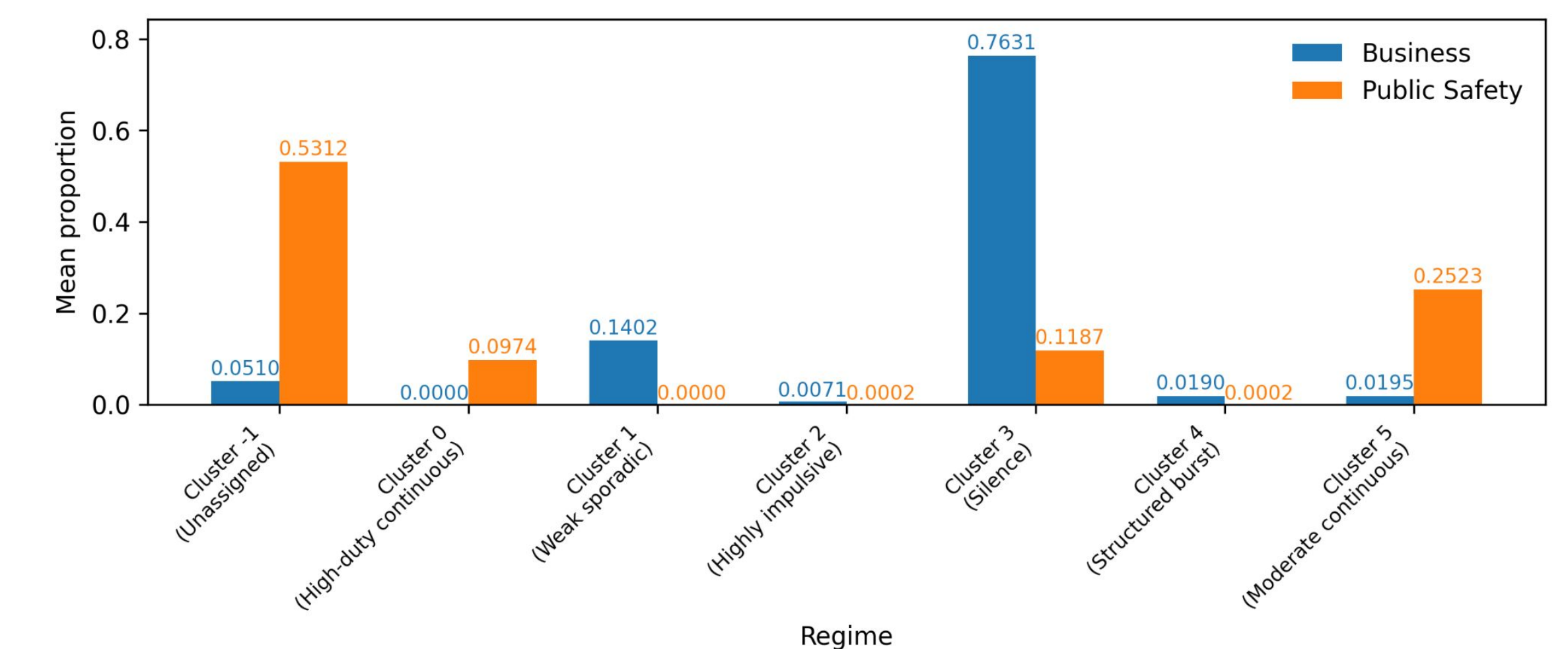


Typical Patterns of Channel Behavior



- Each cluster represents a distinct pattern of channel behavior over time from silence to continuous and burst-dominated activity

Business vs Public Safety: Regime Distribution



- A small number of regimes explains long-term behavior and reveals meaningful differences across channels and services
- Public Safety and Business channels operate in fundamentally different regimes

Our focus:

- One year of data (2020)
- 460–463 MHz LMR band
- 19 licensed channels
- Public Safety + Business use
- No spectral overlap

→ Clean view of real channel behavior

Pipeline

Windows
↓
Features
↓
Clustering
↓
Regimes
↓
Mixtures

Occupancy Hides Structural Differences:

- Left: Duty (occupancy)
- Right: Regime mixture
- Similar-looking occupancy hides different behaviors
- Regime mixtures reveal underlying structural change

Takeaways

- Identify when behavior shifts occur
 - Pinpoint abnormal periods in long-term data
 - Explain changes through interpretable regimes
- Benefits research, operations, and policy